AI Use Cases: Detection and Analysis

AI-powered detection and analysis tools are already embedded in major security platforms. This lesson explores how **signature matching**, **anomaly detection**, **code analysis**, and **fraud detection** work in practice and how platforms like Splunk and Microsoft Sentinel are leading the charge.

LESSON 5.1.2

SECURITY AI

AI in Established Security Platforms

Tools like **Cisco's Splunk** and **Microsoft's Sentinel** have used AI for years, offering a robust suite of capabilities that security teams rely on daily.

Signature Matching

Identify known malicious patterns automatically

Code Quality & Linting

Review and improve organizational code

Vulnerability Assessment

Proactively surface security weaknesses

Anomaly Detection

Flag deviations from normal behavior

Fraud Detection

Identify suspicious financial transactions

Splunk's SPL (Search Processing Language) enables advanced filtering and predictive analytics, while AI can draft detection alerts using natural language processing and query languages.

Signature Matching

Uses **predefined patterns** to identify known malicious activity similar to how antivirus software scans for specific code signatures.

- Matches against a library of known threat signatures
- Generates detailed, actionable alerts automatically
- Can auto-create incident tickets upon detection
- Triggers the appropriate security response workflow

Known Threat Identified

Pattern matched against signature database

Alert Generated

Detailed incident report created automatically

Ticket Auto-Created

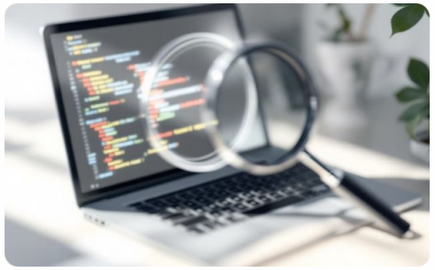
Security team notified and response triggered

Code Quality and Linting Tools

AI-powered linting tools provide security teams with ready-to-use scripts, code completion, and contextual guidance making codebases safer and more maintainable.

Ready-to-Use Scripts

Provide pre-built code scripts and software tools tailored for security team workflows and automation needs.



Code Review & Improvement

Review existing organizational code for quality improvements, offering modification suggestions and readability enhancements.

LLM Security Analysis

Large Language Models analyze code to find security flaws and provide contextual, actionable guidance to developers.

Linting vs. Static Application Security Testing (SAST)

While both tools analyze code, they serve distinct purposes. Understanding the difference helps security teams deploy the right tool for the right job.

🔧 Linting

Focus: Code quality and standards

- Ensures coding standards are consistently met
- Improves readability and maintainability
- Catches syntax errors and style violations
- Helps teams write cleaner, more uniform code

🛡️ SAST

Focus: Security vulnerability detection

- Targets specific security flaws in the codebase
- Detects injection attacks (SQL, command, etc.)
- Identifies cross-site scripting (XSS) vulnerabilities
- Flags broken authentication configurations

Key insight: Linting and SAST are complementary — use linting for ongoing code hygiene and SAST for targeted security audits before deployment.

Anomaly Detection

Anomaly detection evaluates network traffic and communications against **baselines of normal activity**, flagging deviations as potential security incidents.

01

Establish Baseline

Train the AI on what constitutes normal behavior for the specific environment — traffic volumes, access patterns, communication flows.

02

Monitor Continuously

Evaluate real-time network activity and communications against the established behavioral baseline.

03

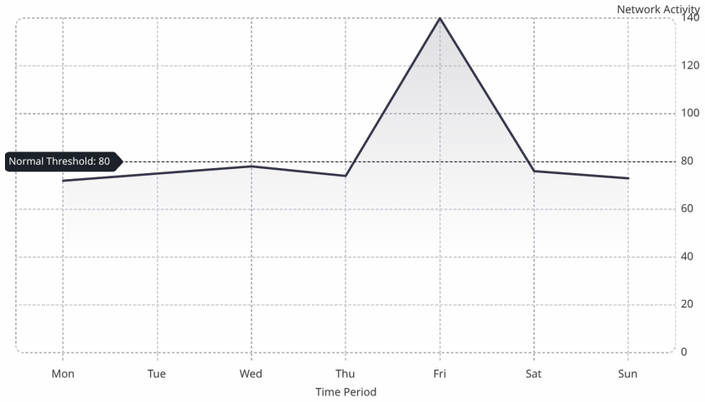
Flag Deviations

Automatically identify and alert on behaviors that deviate from expected norms as potential incidents.

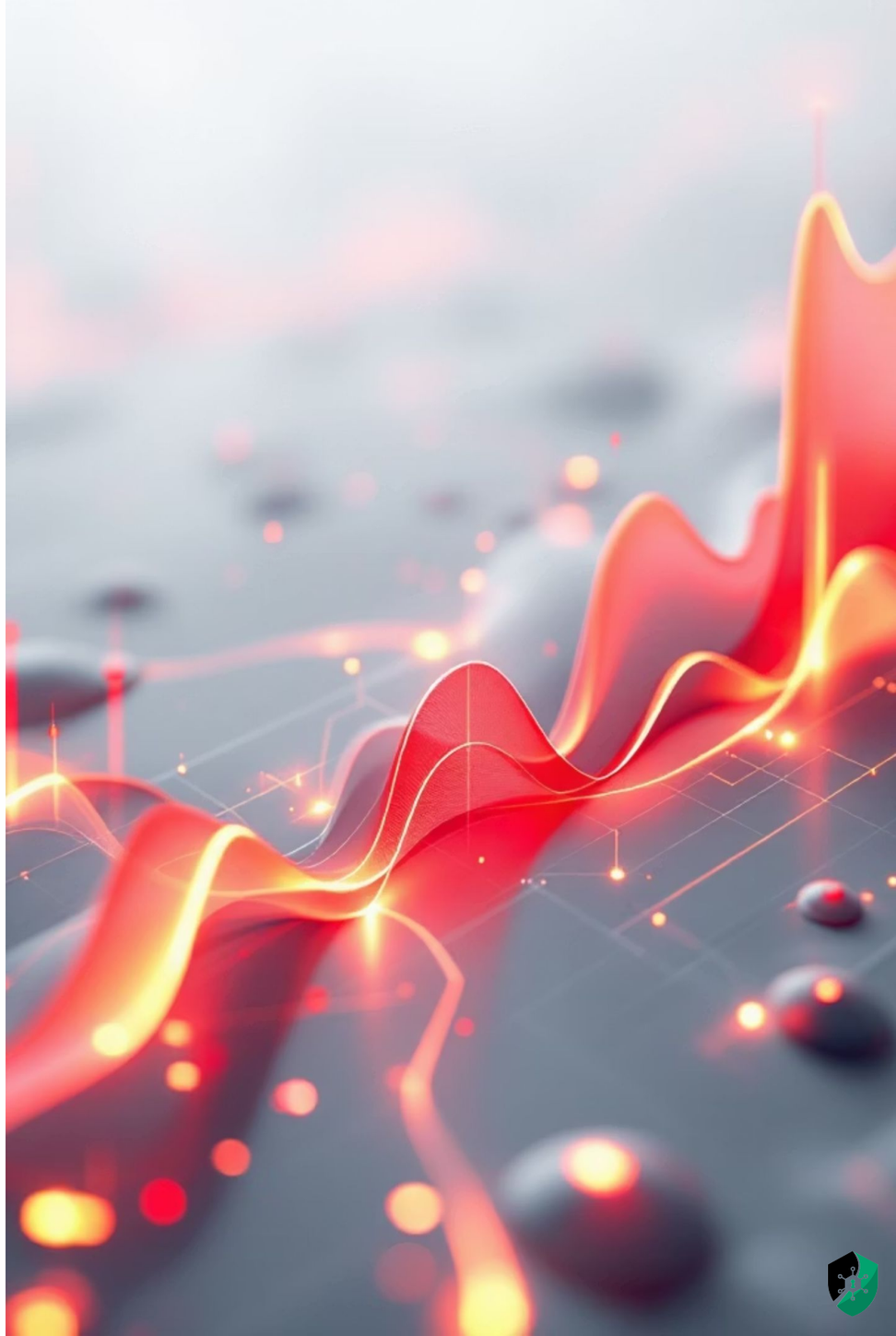
04

Recognize Patterns

Work alongside pattern recognition to identify recurring deviations or emerging trends over time.



A spike above the normal threshold on Friday triggers an anomaly alert for investigation.



Fraud Detection

Fraud detection applies anomaly detection principles to **financial and transactional data**, identifying abnormal data points that signal potential fraud.

1

Collect Transaction Data

Gather historical transaction records including time, location, amount, and merchant category.

2

Build Behavioral Profile

AI learns the cardholder's typical spending patterns, locations, and transaction frequencies.

3

Compare in Real Time

Each new transaction is compared against the profile flagging physically impossible patterns like simultaneous transactions in two countries.

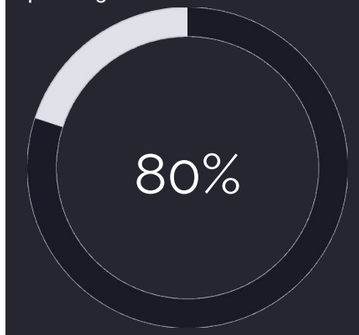
4

Alert & Respond

Suspicious transactions are flagged, blocked, or escalated for human review immediately.

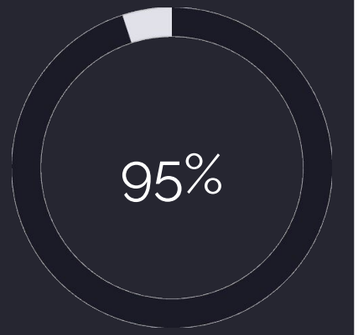
Real-World Example

A credit card is used in New York at 2:00 PM. At 2:15 PM, a transaction appears in Tokyo. The AI flags this as a **physically impossible pattern** — the account is immediately suspended pending verification.



Fraud Reduction

Typical improvement with AI-based detection



Detection Accuracy

In trained financial systems

Key Takeaways

A summary of the core concepts covered in this lesson on AI-powered detection and analysis in cybersecurity.

→ Major Platforms Already Leverage AI

Splunk and Microsoft Sentinel have integrated AI capabilities for years, using tools like SPL for filtering, predictive analytics, and natural language-driven alert drafting.

→ Signature Matching Automates Threat Response

By matching known malicious patterns, signature-based AI tools automatically generate alerts and create incident tickets accelerating the security response cycle.

→ Linting vs. SAST Serve Different Purposes

Linting improves overall code quality and standards compliance, while SAST specifically targets security vulnerabilities like injection attacks, XSS, and broken authentication.

→ Anomaly Detection Identifies Behavioral Deviations

By training on normal baselines, AI flags unusual network traffic and communication patterns working alongside pattern recognition to surface recurring threats.

→ Fraud Detection Applies Anomaly Analysis to Finance

Credit and financial systems use anomaly detection to identify physically impossible transaction patterns, protecting users from fraudulent activity in real time.